

Irrigation Need Prediction - Project Report

This report summarizes a multiclass machine learning project that predicts irrigation requirement (Low, Medium, High) using soil, weather, crop and irrigation related features.

Dataset

10000 samples, 19 input features and 1 target (Irrigation_Need). Target distribution: Low=5864, Medium=3800, High=336. Dataset contains numerical and categorical features.

Features

Soil Type, Soil pH, Soil Moisture, Organic Carbon, Electrical Conductivity, Temperature, Humidity, Rainfall, Sunlight Hours, Wind Speed, Crop Type, Growth Stage, Season, Irrigation Type, Water Source, Field Area, Mulching Used, Previous Irrigation and Region.

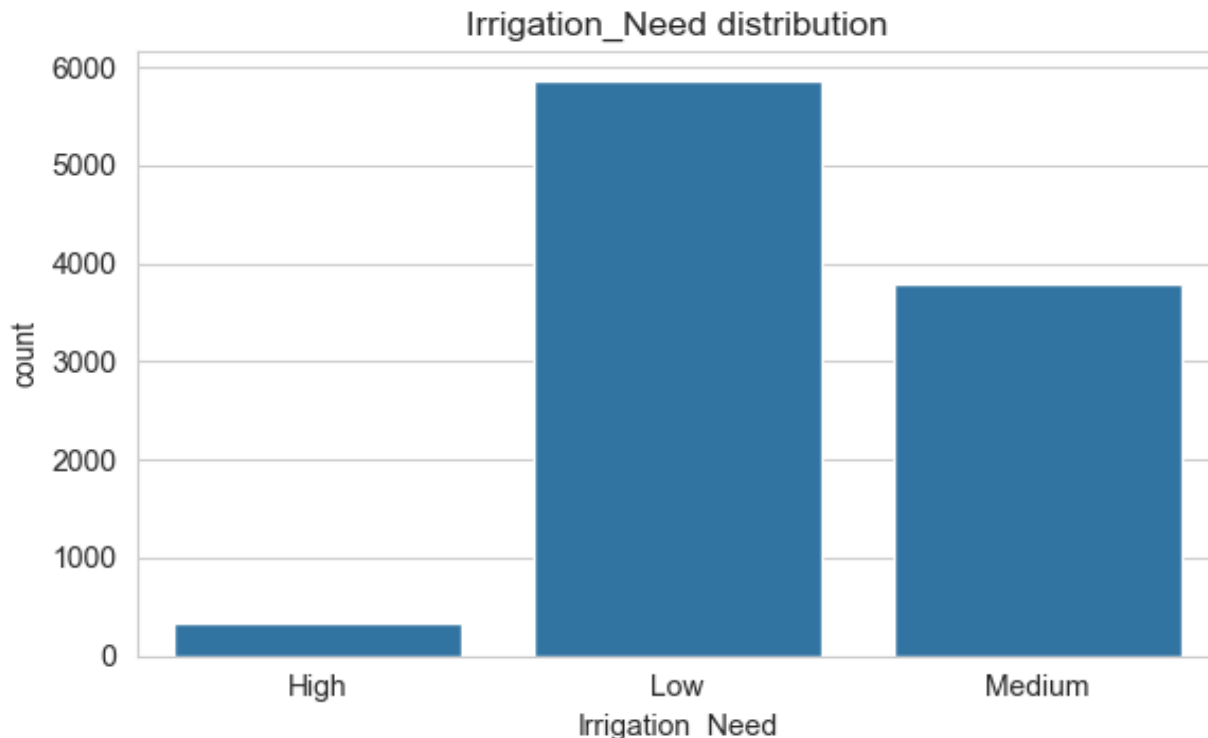
EDA Observations

- Target classes are imbalanced with Low dominating.
- Numeric features are approximately uniformly distributed.
- Correlation between numeric variables is generally weak, indicating low multicollinearity.
- Soil moisture and rainfall together influence irrigation need, though classes overlap.
- Crop growth stage shows noticeable variation across irrigation classes.

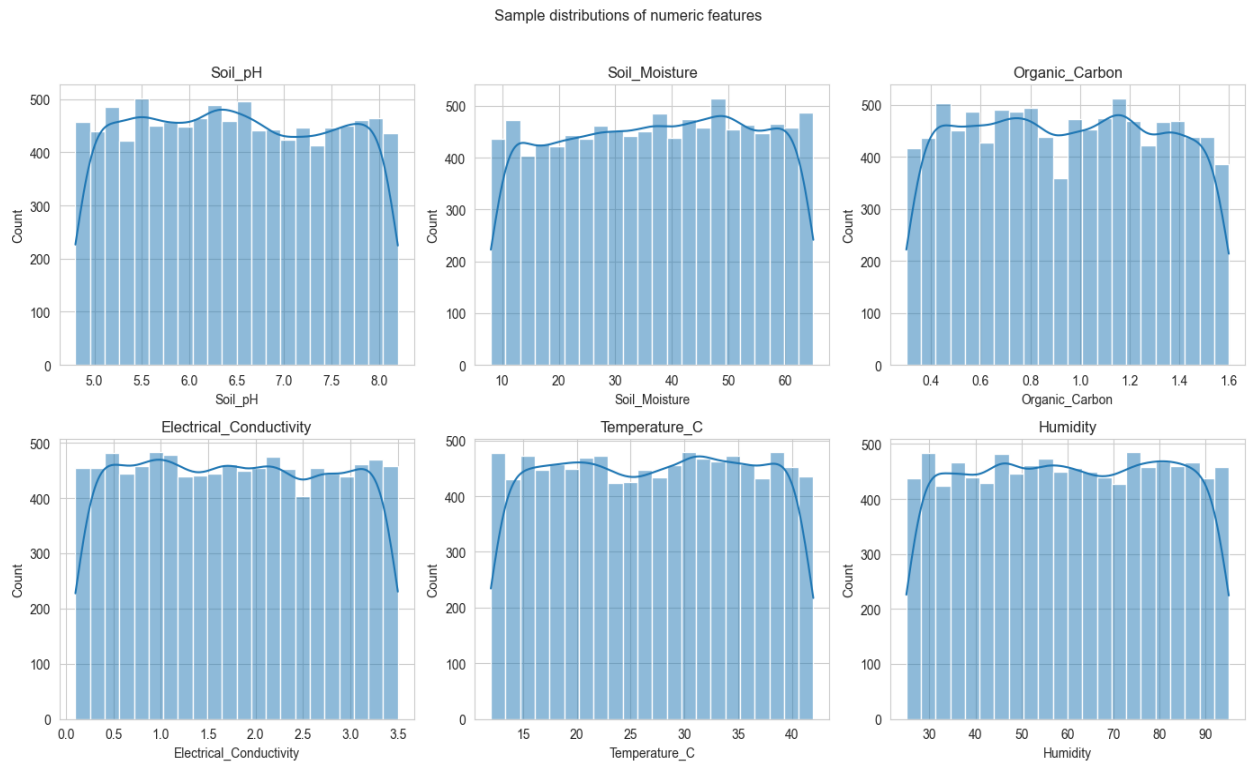
Conclusion

The dataset is suitable for multiclass classification. Class imbalance should be handled using class weights or resampling. Future improvements include hyperparameter tuning, XGBoost, LightGBM and SHAP feature importance.

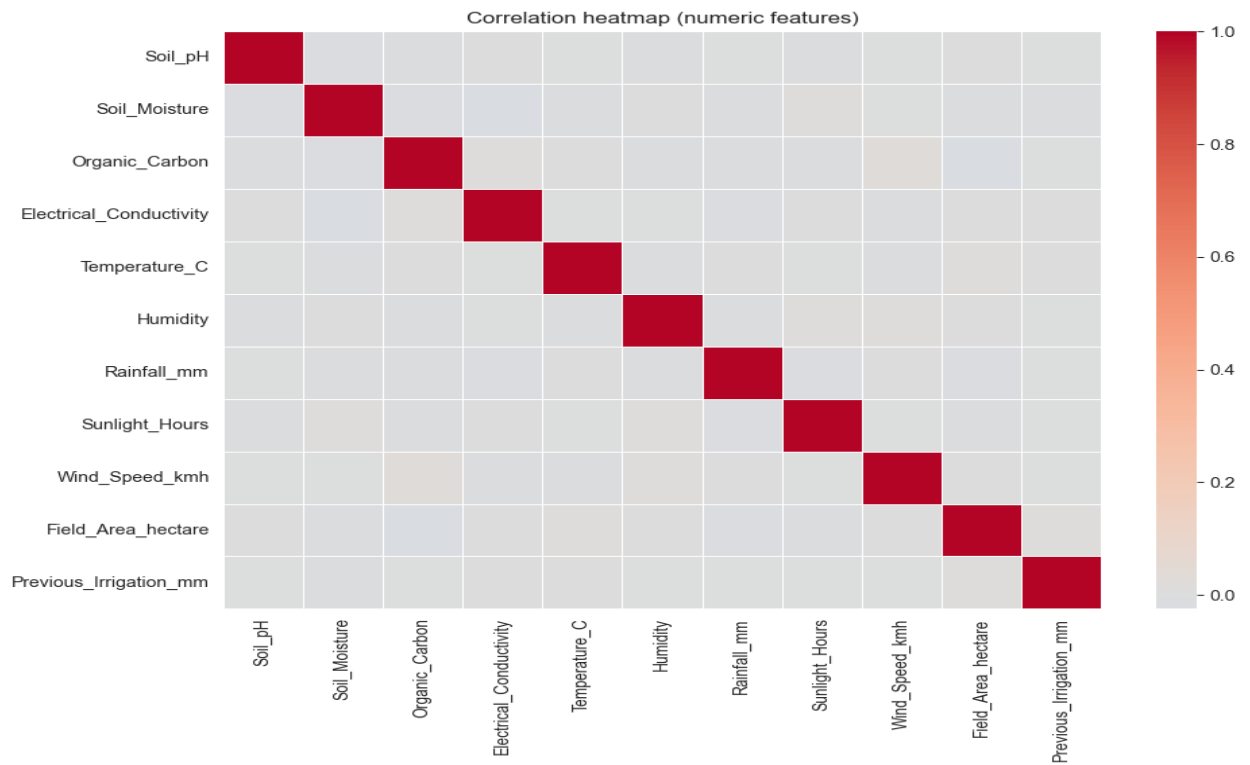
Target Distribution



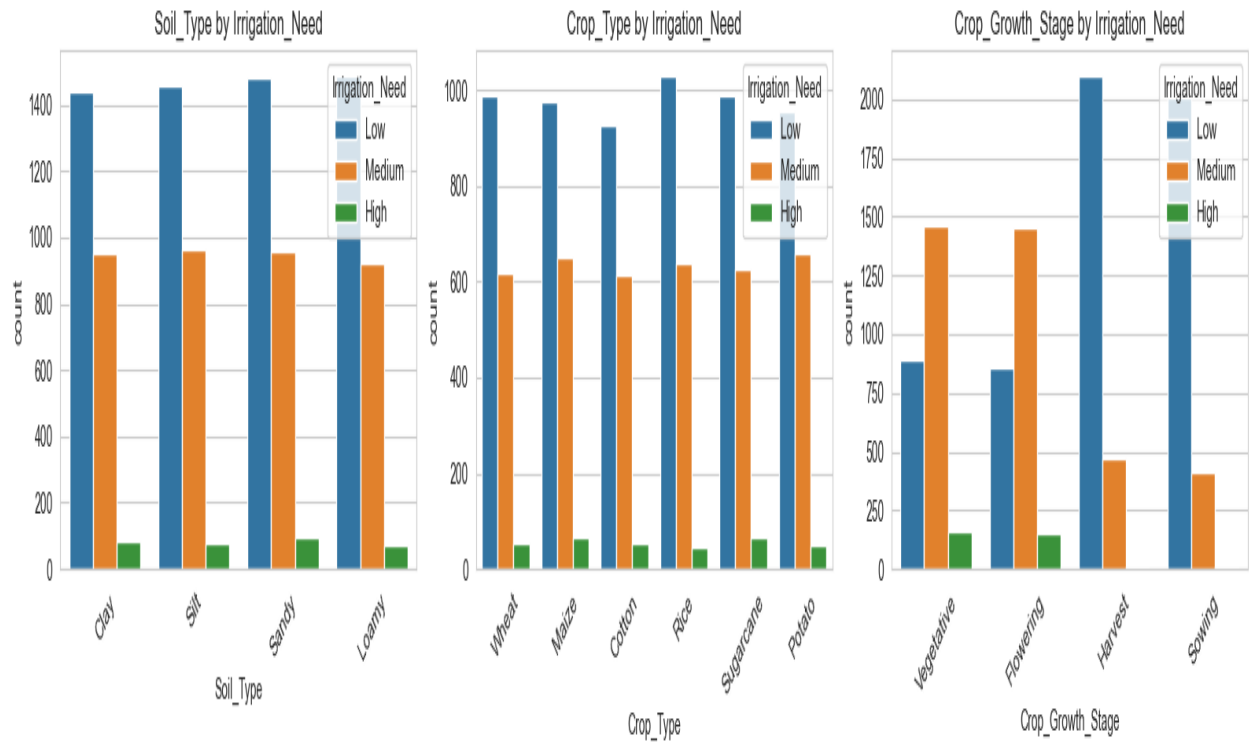
Feature Distributions



Correlation Heatmap



Categorical Feature Analysis



Scatter Plot

